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sleev: An R Package for Semiparametric Likelihood Estimation with Errors in Variables

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Summary

Data with measurement error in the outcome, covariates, or both are not uncommon, particularly with the increased use of routinely collected data for biomedical research. In settings with error-prone data, often only a subsample of study data is validated, also known as two-phase studies. The sieve maximum likelihood estimator (SMLE), which combines the error-prone data on all records with the validated data on a subsample, is a highly efficient and robust estimator to analyze such data. However, given their complexity, a computationally efficient and user-friendly tool is needed to obtain SMLEs. R package `sleev` fills this gap by making semiparametric likelihood-based inference using SMLEs for error-prone two-phase data in settings with binary and continuous outcomes. Functions from this package can be used to analyze data with error-prone responses and covariates.

Statement of Need

Routinely collected data are being used frequently in biomedical research, such as electronic health records. However, these data tend to be error-prone, and careless use of these data could lead to biased estimates and misleading research findings (Duan et al., 2016). To avoid invalid study results, trained experts carefully verify and extract data elements. However, it is usually only feasible to validate data for a subset of records or variables. After validation, researchers have error-prone pre-validation data for all records (phase one data) and error-free validated data on a subset of records (phase two data). Analyses aims to combine the two types of data to obtain estimates that have low bias and are as efficient as possible.

There are some R packages that are effective for data with measurement errors. The R package decon performs nonparametric regression for measurement error models (Wang & Wang, 2011). The R package `simex` and `mecor` approximates the error distribution with different parametric approaches (Lederer & Küchenhoff, 2006; Nab et al., 2021). All of them deals with errors in either outcome or the covariates but not both, and there are no ways to combine the two phases of data with these packages.

SMLE is an estimator that analyzes two-phase data that combines the error-prone data on all records with the validated data on a subsample. SMLE is highly efficient because it utilizes all available data from both phases. (Lotspeich et al., 2022; Tao et al., 2021). SMLE is also robust, since it is not making any parametric assumption on the error model. One of the proofs for the SMLE's robustness is a simulation analysis in Tao et al. (2021), where SMLE is far less biased with error model misspecification when compared with a moment-based estimator. Moreover, SMLE allows model with error-prone outcome and error-prone covariates. Still, in practice these estimators can be difficult to implement, as they involve approximating nuisance conditional

densities using B-splines (Schumaker, 2007) and then maximizing the semiparametric likelihood via a sophisticated EM algorithm (Tao et al., 2017). Here we present R package `sleev` who makes this method readily applicable for practitioners in a user-friendly way (R Core Team, 2024). `sleev` integrates and extends R packages `logreg2ph` and `TwoPhaseReg`, two primitive R packages developed with the original methods papers (Lotspeich et al., 2022; Tao et al., 2021). These two packages lacked proper documentation and were difficult to use. `logreg2ph` was computationally slow. To promote the use of SMLE in two-phase data, extensive work has been done to create `sleev`, a computationally efficient and user-friendly R package to analyze two-phase, error-prone data. Specifically, in `sleev` we rewrote the core algorithms of `logreg2ph` in C++ to speed up the computation (Table 1), and we unified the syntax across functions.

SMLE for linear regression

In this section, we briefly introduce the SMLEs for linear regression. The SMLEs for logistic regression are similar to linear regression and described in the package vignette. Suppose that we want to fit a standard linear regression model for a continuous outcome Y and covariates $\mathbf{X}$: $Y = \alpha + \boldsymbol{\beta}^T\mathbf{X} + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$. Our goal is to obtain estimates of $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}^T, \sigma^2)^T$. When we have error-prone data, Y and $\mathbf{X}$ are unobserved except for a subset of validated records. For the majority of unvalidated records, only the error-prone outcome $Y^* = Y + W$ and covariates $\mathbf{X}^* = \mathbf{X} + \mathbf{U}$ are observed in place of Y and $\mathbf{X}$, where W and $\mathbf{U}$ are the errors for the outcome and covariates, respectively. We assume that $W, \mathbf{U}$ and ϵ are independent. With potential errors in our data, a naive regression analysis using error-prone variables Y^* and $\mathbf{X}^*$ could render misleading results.

We assume that the joint density of the complete data $(Y^*, \mathbf{X}^*, W, \mathbf{U})$ takes the form

$$P(Y^*, \mathbf{X}^*,\ W,\ \mathbf{U}) = P(Y^*|\mathbf{X}^*,\ W,\ \mathbf{U})\, P(W,\ \mathbf{U}|\mathbf{X}^*\ )\, P(\mathbf{X}^*)$$

$$= P_{\boldsymbol{\theta}}\left(Y|X\right) P(W,\ \mathbf{U}|\mathbf{X}^*)\, P(\mathbf{X}^*)$$

where $P(\cdot)$ and $P(\cdot|\cdot)$ denote density and conditional density functions, respectively. $P_{\boldsymbol{\theta}}\left(Y|\mathbf{X}\right)$ then refers to the conditional density function of $Y = \alpha + \boldsymbol{\beta}^T\mathbf{X} + \epsilon$. Denote the validation indicator variable by V, with $V = 1$ indicating that a record was validated and $V = 0$ otherwise. For records with $V = 0$, their measurement errors $(W, \mathbf{U})$ are missing, and therefore their contributions to the log-likelihood can be obtained by integrating out W and $\mathbf{U}$. Let $(Y_i^*, \mathbf{X}_i^*, W_i, \mathbf{U}_i, V_i, Y_i, \mathbf{X}_i)$ for $i = 1, \ldots, n$ denote independent and identically distributed realizations of $(Y^*, \mathbf{X}^*, W, \mathbf{U}, V, Y, \mathbf{X})$ in a sample of n subjects. Then, the observed-data log-likelihood takes the form

$$\sum_{i=1}^{n} V_i\{\log P_{\boldsymbol{\theta}}\left(Y_i|\mathbf{X}_i\right) + \log P(W_i, \mathbf{U}_i|\mathbf{X}_i^*)\}$$

$$+ \sum_{i=1}^{n} (1 - V_i) \log\left\{\int\int P_{\boldsymbol{\theta}}\left(Y_i^* - w|\mathbf{X}_i^* - \mathbf{u}\right) P(w, \mathbf{u}|\mathbf{X}_i^*)\, dw d\mathbf{u}\right\} \qquad (1)$$

where $P(\boldsymbol{X^*})$ is left out, because the error-prone covariates are fully observed and thus $P(\boldsymbol{X^*})$ can simply be estimated empirically. We estimate the unknown measurement error model, $P(W_i, \mathbf{U}_i|\mathbf{X}_i^*)$ using B-spline sieves. Specifically, we approximate $P(w_i, \mathbf{u}_i|\mathbf{X}_i^*)$ and $\log P(W_i, \mathbf{U}_i|\mathbf{X}_i^*)$ by $\sum_{k=1}^{m} I\left(w_i = w_k, \mathbf{u}_i = \mathbf{u}_k\right) \sum_{j=1}^{s_n} B_j^q\left(\mathbf{X}_i^*\right) p_{kj}$ and $\sum_{k=1}^{m} I\left(W_i = w_k, \mathbf{U}_i = \mathbf{u}_k\right) \sum_{j=1}^{s_n} B_j^q\left(\mathbf{X}_i^*\right) \log p_{kj}$, respectively. $\{(w_1, \mathbf{u}_1), \ldots, (w_m, \mathbf{u}_m)\}$ are the m distinct observed $(W, \mathbf{U})$ values, $B_j^q\left(\mathbf{X}_i^*\right)$ is the jth B-spline basis function

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of order q evaluated at $\mathbf{X}_i^*$, s_n is the dimension of the B-spline basis, and p_{kj} is the coefficient associated with $B_j^q(\mathbf{X}_i^*)$ and $(w_k, \mathbf{u}_k)$. The log-likelihood in expression (1) is now approximated by

$$\sum_{i=1}^{n} V_i \left\{ \log P_{\boldsymbol{\theta}}(Y_i|\mathbf{X}_i) + \sum_{k=1}^{m} \{ I(W_i = w_k, \mathbf{U}_i = \mathbf{u}_k) \sum_{j=1}^{s_n} B_j^q(\mathbf{X}_i^*) \log p_{kj} \} \right\}$$

$$+ \sum_{i=1}^{n} (1 - V_i) \log \left\{ \sum_{k=1}^{m} \{ P_{\boldsymbol{\theta}}(Y_i^* - w_k | \mathbf{X}_i^* - \mathbf{u}_k) \sum_{j=1}^{s_n} B_j^q(\mathbf{X}_i^*) \log p_{kj} \} \right\}. \qquad (2)$$

The maximization of expression (2) is carried out through an EM algorithm to find the SMLEs $\hat{\boldsymbol{\theta}}$ and $\hat{p}_{kj}$. The covariance matrix of the SMLE $\hat{\boldsymbol{\theta}}$ is obtained through the method of profile likelihood (Murphy & Van der Vaart, 2000). Full details on the SMLE method for logistic regression with error-prone data, including its theoretical properties, can be found in (Lotspeich et al., 2022).

Functionalities of the `sleev` R package

The `sleev` package provides a user-friendly way to obtain the SMLEs and their standard errors. The package can be installed from CRAN or GitHub. The `sleev` package includes two main functions: linear2ph() and logistic2ph(), to fit linear and logistic regressions, respectively, under two-phase sampling with an error-prone outcome and covariates. The input arguments are similar for the two functions and listed in Table 2. From Table 2, we see that in addition to the arguments for error-prone and error-free outcome and covariates, the user needs to specify the B-spline matrix $B_j^q(\mathbf{X}_i^*)$ to be used in the estimation of the error densities.

The `sleev` package included a dataset constructed to mimic data from the Vanderbilt Comprehensive Care Clinic (VCCC) patient records from (Giganti et al., 2020). Table 3 lists the variables in this dataset to be used in subsequent analyses.

Example: Case study with mock data

We now illustrate how to obtain SMLEs using the `sleev` package with dataset `mock.vccc`. Specifically, we show how to fit a linear regression model in the presence of errors in both the outcome and covariates using the linear2ph() function. Situations with more covariates and examples with logistic regression are included in the package vignette.

This example fits a linear regression model with CD4 count at antiretroviral therapy (ART) initiation regressed on viral load (VL) at ART initiation, adjusting for sex at birth. Both CD4 and VL are error-prone, partially validated variables, whereas sex is error-free.

Because of skewness, we often transform both CD4 and VL. In our analysis, CD4 was divided by 10 and square root transformed and VL was $\log_{10}$ transformed:

```
library("sleev")
data("mock.vccc")
mock.vccc$CD4_val_sq10 <- sqrt(mock.vccc$CD4_val / 10)
mock.vccc$CD4_unval_sq10 <- sqrt(mock.vccc$CD4_unval / 10)
mock.vccc$VL_val_l10 <- log10(mock.vccc$VL_val)
mock.vccc$VL_unval_l10 <- log10(mock.vccc$VL_unval)
```

To obtain the SMLEs, we first need to set up the B-spline basis for the covariates `VL_unval_l10` (the transformed error-prone VL variable from phase one). The `spline2ph()` function in `sleev` packages can set up the B-spline basis, and combine it with the data input for the final analysis.

Here, we use a cubic B-spline basis with the degree=3 argument. The size of the basis s_n is set to be 20, specified through the `size = 20` argument. The B-spline basis is set up separately for the two Sex groups, and the size of the B-spline basis is assigned in proportion to the relative size of the two Sex groups. This is specified by argument `group`. This allows the errors in `VL_unval_l10` to be heterogeneous between males and females. The described B-spline basis is constructed as follows.

```
sn=20
data.linear <- spline2ph(x = "VL_unval_l10", data = mock.vccc, size = sn,
                         degree = 3,  group = "Sex")
```

Alternatively, if the investigator has prior knowledge that the errors in `VL_unval_l10` are likely to be homogeneous, one may fit a simpler model by not stratifying the B-spline basis by Sex.

The SMLEs can be obtained by running function `linear2ph()`, as shown in the code below. The fitted SMLEs are stored in a list object of class `linear2ph`. Here, we assign the fitted SMLEs to the variable name `res_linear`. The list of class `linear2ph` contains five components: `coefficient`, `covariance`, `sigma`, `converge`, and `converge_cov`.

```
start.time <- Sys.time()
res_linear <- linear2ph(y_unval = "CD4_unval_sq10", y = "CD4_val_sq10",
                        x_unval = "VL_unval_l10", x = "VL_val_l10", z = "Sex",
                        b_spline = colnames(Bspline), data = data.linear,
                        hn_scale = 1, se = TRUE, tol = 1e-04,
                        max_iter = 1000, verbose = FALSE)
paste0("Run time: ", round(difftime(Sys.time(), start.time,
                                    units = "secs"), 3), " sec")

[1] "Run time: 1.78 sec"
```

We should first check if the EM algorithms for estimating the regression coefficients and their covariance matrix converged by the `print()` for class `linear2ph` directly:

```
> res_linear
This model has converged.
Coefficients:
 Intercept VL_val_l10        Sex
     4.821    -0.141      0.273
```

The `summary()` function for list of class `linear2ph()` returns the coefficient matrix and covariance matrix as follows:

```
> summary(res_linear)
Model Summary:
Coefficients:
            Estimate     SE Statistic  p-value
Intercept      4.821 0.1587     30.39 0.000000
VL_val_l10    -0.141 0.0398     -3.55 0.000389
Sex            0.273 0.1089      2.51 0.012229

Covariance Matrix:
          [,1]      [,2]      [,3]
[1,]  0.02517 -5.06e-03 -8.87e-03
[2,] -0.00506  1.59e-03 -5.84e-05
[3,] -0.00887 -5.84e-05  1.19e-02
```

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Table 1: Logistic regression function computational time (s) comparison, across 100 simulations.

Functions	Mean	SD
`logreg2ph`	155.814571	72.41892927
`logistic2ph`	70.569576	8.38126354

Table 2: Main arguments for linear2ph() and logistic2ph()

Argument	Description
Y_unval	Column name of unvalidated outcome in the input dataset.
Y	Column name of validated outcome in the input dataset. NAs in the input will be counted as individuals not selected in phase two.
X_unval	Column names of unvalidated covariates in the input dataset.
X	Column names of validated covariates in the input dataset. NAs in the input will be counted as individuals not selected in phase two.
Z	Column names of error-free covariates in the input dataset.
Bspline	Column names of the B-spline basis in the input dataset.
data	Dataset
hn_scale	Scale of the perturbation constant in the variance estimation via the method of profile likelihood. The default is 1.
noSE	Standard errors of the parameter estimators will not be estimated when set to TRUE. The default is FALSE.
TOL Convergence criterion.	The default is 0.0001.
MAX_ITER	Maximum number of iterations in the EM algorithm. The default is 1000.
verbose	Print analysis details when set to TRUE. The default is FALSE.

Table 3: Data dictionary for mock.vccc

Name	Status	Type	Description
ID	error-free		Patient ID
VL_unval	error-prone	continuous	Viral load (VL) at antiretroviral therapy (ART) initiation
VL_val	validated		
ADE_unval	error-prone	binary	Had an AIDS-defining event (ADE) within one year of ART initiation: 1 - yes, 0 – no
ADE_val	validated		
CD4_unval	error-prone	continuous	CD4 count at ART initiation

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Name	Status	Type	Description
CD4_val	validated		
prior_ART	error-free	binary	Experienced ART before enrollment: 1 - yes, 0 - no
Sex	error-free	binary	Sex at birth of patient: 1 - male, 0 - female
Age	error-free	continuous	Age of patient

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